

Emergency Scheduling for Forest Fires Subject to Limited Rescue Team Resources and Priority Disaster Areas

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To enable immediate and efficient emergency scheduling during forest fires, we propose a novel emergency scheduling model for such fires subject to priority disaster areas and limited rescue team resources to minimize the total travel distance for rescue teams. Moreover, a hybrid intelligent algorithm integrating genetic algorithm (GA) and particle swarm optimization (PSO) is adopted to solve the proposed model. A case study is presented to illustrate the proposed model and the effectiveness of the proposed algorithm. The goal of this work is to analyze the emergency scheduling problem of forest fires subject to limited rescue teams and priority disaster areas. Both theoretical and simulation results demonstrate that the proposed model can perform effectively the quantitative analysis of an emergency involving forest fires. Such results can help decision makers to make better judgment when dealing with an emergency involving fires. © 2016 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

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1. Introduction

Emergency management (EM) is concerned with information collection, dispatch, transportation, damage, time response, and material resources, as well as accurately tracking the disaster-hit area in all kinds of emergency events [1]. Since its initiation, there have been many advances in their solution methods, variants, and applications, in, e.g. earthquake relief, traffic incident response, personnel evacuation, flood emergency, and forest fires. Many different models have been formed, e.g. integer linear programming models [2], dynamic programming models [3], goal programming models [4], hierarchical compromise model [5], and game model [6]. Meanwhile, a large number of approaches for solving the different models have been proposed, e.g. meta-heuristic algorithm [7] genetic algorithm (GA) [8], immune algorithm [9], branch-and-cut algorithm [10], self-organizing map approach [11], and greedy algorithm [12].

Route scheduling is one of the fundamental problems of EM and is attracting more and more attention of researchers [13–15]. For instance, Ozdamar *et al.* constructed a vehicle routing model in emergency logistics for natural disasters [16]. Zhang *et al.* presented an optimal path model to minimize the travel distance in an emergency logistics network [17]. Yuan and Wang established a single-objective path selection model with the minimum response time and used the Dijkstra algorithm to solve it [18]. Chang *et al.* proposed an optimal model to minimize the travel time, resource demand, and transport cost with resource constraints, and used a greedy-search-based GA to solve it [19]. Wilson *et al.* described a path-scheduling model for medical emergency service during mass casualty incidents [20]. Yi and Kumar adopted an ant colony optimization method to solve a logistics scheduling problem arising

during disaster relief activities [21]. Barbarosoglu and Arda presented a two-stage stochastic programming framework for transportation planning during disaster response [22]. Wang *et al.* presented a decision support system for emergency scheduling [23]. Ozdamar and Pdamallu compared two mathematical models for earthquake relief logistics [24].

The above overview suggests that the current EM research mainly focuses on the analysis of travel time or distance for a single disaster area with one or more rescue teams, i.e. each disaster area can be reached by one or more rescue teams in an emergency activity without much consideration of the rescue team accounts. However, sometimes rescue teams will be limited, and decision makers cannot ensure that each disaster area has a rescue team arriving in the first time, i.e. one rescue team probably needs to reach one area and then another area. Moreover, in some emergency events, there are two or more disaster areas and their situations may be different, i.e. the disaster situation of some areas is more severe than in other areas, and it is necessary to take priority order among different disaster areas into account when dealing with route scheduling problem for EM. Thus we must address a route scheduling problem for EM with limited rescue team resources and priority disaster areas. In addition, natural hazards frequently occur around the world, and forest fires are one of the common natural hazards, raising forest fire management problems. For example, Bancal and Pastor-Satorras presented a dynamic model of forest fire mechanism [25]. Iudin *et al.* presented an infinity computation of a forest fire model [26]. Margerit and Séro-Guillaume established a propagation model for forest fires [27]. Aditama and Dennekamp discussed the impact of haze from forest fire on respiratory health [28,29]. Although some researchers have proposed the mechanism of forest fires, they have paid little attention to the emergency scheduling of forest fires. To do so, this work addresses an emergency scheduling of forest fires with limited rescue (fire fighting) team resources and priority disaster areas/fire points, for the first time to the best our knowledge. The aim of this work is, therefore, to find a new way to determine the optimal scheduling program of forest fires by taking priority fire points and the limited fire fighting teams into

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full account. Additionally, the scheduling or planning problems are almost NP-hard, and for our problem there are a set of scheduling solutions subject to limited rescue team resources and a specific rescue priority, which is rather hard and inconvenient to be solved via a mathematical programming approach or an exact method, e.g. CPLEX. Thus, a hybrid heuristic algorithm integrating GA and particle swarm optimization (PSO) is adopted to solve the proposed problem. Compared with the existing work [8,14], we make contributions: (i) In order to describe the actual situation of emergency scheduling for forest fires, we propose an optimization model of rescue scheduling for forest fires considering limited fire fighting team resources and the rescue order of disaster areas to minimize the travel distance of fire fighting teams under their specific travel time constraints; (ii) To solve the optimization problem, we formulate a hybrid method integrating PSO and GA. By comparing with GA and PSO, this work validates the effectiveness and feasibility of the proposed model in solving this scheduling optimization problem.

In Section 2, an emergency scheduling problem for forest fires is described and its emergency scheduling mathematical model is established. Section 3 introduces the hybrid algorithm to solve the proposed model. In Section 4, an actual example is presented to test the effectiveness of the proposed method. Finally, Section 5 concludes our work and describes some future research issues.

2. Problem Statement

This work proposes a new emergency scheduling problem for forest fires involved with limited fire fighting team resources and multiple fire areas/points. Its aim is to determine an optimal feasible emergency schedule of forest fires such that fire fighting teams should be dispatched to different fire points to minimize the total travel distance of the fire fighting teams while meeting the specific time constraints and the priority order requirement of the fire points. Note that a fire fighting team is allowed to reach other fire points after extinguishing a fire point. Also, a priority level is introduced to represent the severity of fire situation of each fire point, and it is denoted as P_i , $i = 1, 2, \dots, N$, where i is the index of fire points and N represents the number of all points. Namely, the more severe the fire situation, the higher priority level and the larger the P_i , which means the corresponding fire point has the priority over the other points. Hence a fire team should try to arrive at this priority fire point preferentially. We call this problem an emergency scheduling problem for forest fires subject to limited fire fighting team resources and priority disaster areas/fire points.

In order to conveniently describe our problem, the following notations and parameters are defined and assumptions are made.

2.1. Parameters The notations are summarized as follows:

1. m : The index of a fire fighting team, $m = 1, 2, \dots, M$, where M is the number of fire fighting teams.
2. i, j : The index of a point/area, $i, j = 0, 1, 2, \dots, N$, where N represents the number of all points. Note that 0 represents the starting point and others represent fire points.
3. V_m : The velocity of the m th fire fighting team, i.e. average vehicle velocity of the m th fire fighting team.
4. d_{ij} : The travel distance between points i and j .
5. P_i : Priority level of the i th fire point; it represents the severity of fire situation of i th fire point.
6. $y_{ij}^m = 1$, if the m th fire fighting team goes to point j from point i ; otherwise, $y_{ij}^m = 0$.

2.2. Assumptions We make the following assumptions:

1. The fire extinguishing material is sufficient to meet the emergency requirements.

2. The model is suited to a small/medium-sized forest fire.
3. The number of fire fighting teams is known; also the priority level of each fire point is known.
4. The fire fighting teams' resources are limited. That is, the number of fire points N is larger than the number of fire fighting teams M , i.e. $N > M$.

2.3. Mathematical model By considering the goal of minimum total travel distance of fire fighting teams, we formulate an emergency scheduling model for forest fires subject to limited fire fighting teams' resources and priority disaster areas (fire points) as follows:

$$\text{Min } D = \sum_{m=1}^M \sum_{i=1}^N \sum_{j=1}^N d_{ij} y_{ij}^m + \sum_{m=1}^M \sum_{j=1}^N d_{0j} y_{0j}^m + \sum_{m=1}^M \sum_{i=1}^N d_{i0} y_{i0}^m \quad (1)$$

$$\text{s.t. } \sum_{i=0}^N \sum_{j=0}^N \frac{d_{ij}}{V_m} \leq T'_m, \quad m = 1, 2, \dots, M \quad (2)$$

$$\sum_{i=0}^N \sum_{m=1}^M y_{ij}^m = 1, j = 1, 2, \dots, N \quad (3)$$

$$\sum_{i=0}^N y_{ij}^m - \sum_{i=0}^N y_{ij}^m = 0, m = 1, 2, \dots, M, j = 0, 1, 2, \dots, N \quad (4)$$

$$P_i \geq P_j, i, j = 1, 2, \dots, N \quad (5)$$

$$\forall P_i \geq P_j, \sum_{i=1}^N \sum_{m=1}^M y_{0i}^m = M \text{ and}$$

$$\sum_{j=1}^N \sum_{m=1}^M y_{0i}^m = N - M, \quad (6)$$

$$i, j = 1, 2, \dots, N, N > M$$

$$\sum_{i=1}^N \sum_{j=1}^N y_{ij}^m \geq 0, m = 1, 2, \dots, M \quad (7)$$

$$y_{ij}^m \in \{0, 1\}, m = 1, 2, \dots, M, i, j = 0, 1, 2, \dots, N \quad (8)$$

Objective function (1) represents the minimum travel distance of the fire fighting teams in an emergency situation, where its first item is the sum of the travel distances of the fire fighting teams between the fire points; the second one is the sum of the travel distance of each fire team from the starting point to the first-arrival fire point; the last one is the sum of the travel distance of each fire team from the last-arrival fire point to the starting point. Constraint (2) represents a travel time constraint to schedule a fire fighting team, where T'_m is the given time to schedule a fire fighting team; constraint (3) stipulates that a fire point must be covered only once by a fire fighting team; constraint (4) represents the equilibrium constraint of in-degree and out-degree of a fire point, i.e., a fire fighting team should leave a fire point after the fire that point is extinguished; constraint (5) represents the priority level constraint of each fire point; constraint (6) represents M fire fighting teams should first reach the M higher priority fire points because of the limited fire fighting team resources, and $N - M$ lower priority ones are rescued subsequently; constraint (7) represents each fire fighting team should put the fire out at least at one point; and constraint (8) represents that the decision variables can take 1 or 0.

3. Hybrid Algorithm

GA and PSO as heuristic algorithms are used to successfully solve many complex industrial optimization problems because of their strong optimization ability [30–34]. In order to make full

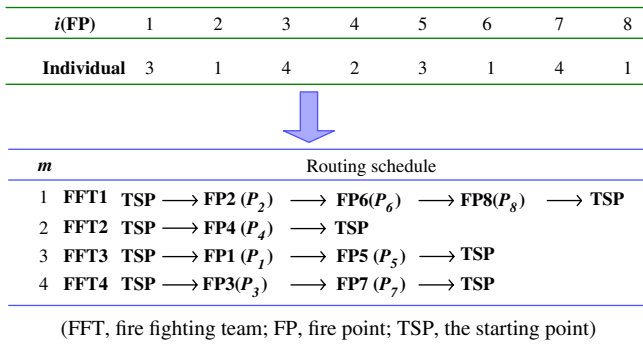


Fig. 1. Example mapping for an individual transforming into a routing schedule

use of the global optimization ability of GA and the fast convergence ability of PSO, a hybrid intelligent algorithm integrating GA and PSO is adopted to solve the proposed problem, and PSO of global and local search is introduced into GA. The hybrid algorithm mainly contains five tasks: generating diversified initial individuals, calculating the fitness value of the objective function, searching for the personal best and global best, crossover, mutation, and updating the individual.

3.1. Initialization The encoding form of a solution directly affects the efficiency of the proposed algorithm. According to the nature of the problem, the integer encoding structure is adopted as the encoding form, in which an individual is expressed by an array with n elements, in which the element value represents the serial number of fire fighting teams, which is randomly numbered from 1 to M , and in which the index of an array represents the serial number of fire points, which is arranged from high to low according to the priority level P_i of each fire point and is numbered from 1 to N . In addition, an array is divided into two sections and they are randomly initialized. The first section contains M elements which denote M higher priority fire points subject to constraints 5 and 6, and the second section contains $N-M$ elements which denote the other $N-M$ lower priority fire points subject to constraint 5. For example, let us randomly select an individual $I = \{3, 1, 4, 2, 3, 1, 4, 1\}$, where $M = 4$, $N = 8$, and transform it into a routing schedule as shown in Fig. 1. Note that the priority level of each fire point is $P_1 > P_2 > P_3 > P_4 > P_5 > P_6 > P_7 > P_8$.

The initial process for the emergency routing scheduling problem of a forest fire is as follows:

Usually, as the number of fire point increases, the number of path solutions also increases drastically. Clearly, some solutions/individuals among them are infeasible, that is, they cannot meet specific constraints, e.g. the time constraint. They need to be adjusted and a common method to do so is regeneration or reconstruction. We adopt the first one in this work. Its procedure is presented in Algorithm 1.

3.2. Assessment of the objective function The objective function contains three terms, which are adopted to evaluate the objective fitness value of the corresponding individual. The smaller the value, the better the individual, i.e. the corresponding individual is more likely to be preserved. The fitness value of the objective function can be obtained from (1).

3.3. Searching for the optimal solutions, personal and global The personal best refers to searching for the best solution in a specific location where the individual has experience, i.e. optimal solution of individual (OSI). Global best

Algorithm 1 The generation method of initial path set Q

Input : I

Output : Q

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(1)   $i = 1$ 
(2)  While ( $i \leq \text{PopSize}$ ) Do
(3)      Flag=0;
(4)      While (Flag=0) Do
(5)          Randomly generate the first section of coding subject to
              constraints 5 and 6;
(6)          Randomly generate the second section of coding subject
              to constraint 5;
(7)          Construct the initial solution  $I$ ;
(8)          If ( $I$  is feasible, i.e., meeting constraint 2)Then
(9)              Insert  $I$  to the initial solution set  $Q$ ;
(10)         Flag=1;
(11)         Else
(12)             Flag=0;
(13)         End While;
(14)          $i++$ ;
(15)     End While;
```

refers to searching for the best solution in each location where all individuals have experience, i.e. optimal solution of population (OSP). The OSI and OSP are used to store and update high-quality and diversified solutions. Such solutions are selected from the current population Q and the current individual (CI) that have experienced crossover and mutation operations. After calculating the fitness value of CI, local optimization is performed by OSI comparing with CI, and personal optimal solution is better for both. Global optimization is performed by OSP comparing with CI, and global optimal solution is better for both.

3.4. Crossover and updating the individual The offspring individual is formed by combining CI and OSP. As initial population, we still divide the offspring individual into two sections. Its crossover process is presented as follows: the former section is carried out by a double-point crossover method, and then the second section is performed in the same way. Compared to other crossover combination operators, this method has the advantage that the priority constraints of each individual of parents are completely enforced from one generation to the next and it is ensured that each individual of the offspring meets priority constraints. The crossover operation is described in Algorithm 2.

3.5. Mutation and updating the individual Mutation operation is similar to crossover operation, i.e. the former section is carried out by an integer mutation method, and then the second section is performed by the same way. Moreover, the mutation of each individual must be performed in the proposed method, which ensures better diversity of the population. The mutation process of the proposed algorithm is given in Algorithm 3.

3.6. Procedure of proposed hybrid algorithm Through the above steps, an offspring individual is updated only if it is feasible and its corresponding objective function value is better than that of the parent. By repeating this process a given PopSize times, a new population is obtained. Its optimal solution is evaluated and updated by the searching personnel and global optimal solutions. In addition, the proposed algorithm adopts the maximum iteration count MaxIter as the termination criterion, i.e. when the maximum iteration count is reached, the algorithm is terminated. The proposed algorithm is shown in Fig. 2.

Table I. The travel distance of each fire point

Point no. (N)	0	1	2	3	4	5	6	7
0	0	42	56	63	65	50	66	45
1	42	0	20	33	45	35	48	64
2	56	20	0	46	53	44	42	58
3	63	33	46	0	60	55	52	54
4	65	45	53	60	0	62	64	56
5	50	35	44	55	62	0	65	66
6	66	48	42	52	64	65	0	63
7	45	64	58	54	56	66	63	0

Table II. Optimal scheme results

m	V_m	T'_m	D_m	$\frac{D_m}{V_m}$	D	Quasi-optimal solution
1	54	3.5	181	$3.35 \leq 3.5$	519	{2, 3, 1, 2, 3, 1, 2}
2	54	3.5	188	$3.48 \leq 3.5$		
3	54	3.5	150	$2.78 \leq 3.5$		

Table III. Optimal scheme results with 5.5 h time constraints

m	V_m	T'_m	D_m	$\frac{D_m}{V_m}$	D
1	40	5.5	210	$5.25 \leq 5.5$	518
2	40	5.5	181	$4.53 \leq 5.5$	
3	40	5.5	127	$3.18 \leq 5.5$	

Table IV. Optimal scheme results with 10 h time constraints

m	V_m	T'_m	D_m	$\frac{D_m}{V_m}$	D
1	40	10	224	$5.60 \leq 10$	515
2	40	10	127	$3.18 \leq 10$	
3	40	10	164	$4.10 \leq 10$	

$MaxIter = 100$. Additionally, the algorithm runs on an Intel(R) Core(TM) i3 CPU (2.53 GHz/4.00 G RAM) PC with the Windows 7 operating system.

After the algorithm is executed, the following results were obtained: the optimal individual is {2, 3, 1, 2, 3, 1, 2} and the minimum travel distance of fire fighting teams is 519 km. It denotes that the optimal scheme is one in which fire fighting team 1 needs to extinguish fire points 3 and 6; fire fighting team 2 needs to extinguish fire points 1, 4, and 7; and fire fighting team 3 needs to extinguish fire points 2 and 5. In addition, the optimal scheme is shown in Table II. Note that D_m denotes the total travel distance of the m th fire fighting team, and T'_m is generally determined by the practical fire condition and the number of rescue team resources.

To test the effectiveness of the proposed model, the optimal scheduling results with given different time constraints are presented in Tables III and IV, respectively.

From Tables III and IV, given different transportation time constraints, different optimal travel distances are obtained. The

results show that the proposed model can effectively express and achieve the time–distance tradeoff.

In order to test the effectiveness of the algorithm, the results of the case study with different run times and algorithm parameters are shown in Table V. Their relative errors were computed, and are shown in Table V and the relative error is defined as $(\text{actual value} - V^*)/V^* \times 100\%$, where V^* denotes the experimentally obtained best value, i.e. the minimum value of solution results of the model with different run times. The actual values are the solution results of the model when the algorithm is run each time.

From Table V, the relative error does not exceed 1.35%. This indicates that the proposed algorithm is highly satisfactory when used to solve the proposed models.

In addition, in order to further test the effectiveness and rationality of the algorithm, its efficiency and that of the existing GA [8] and PSO [36] algorithms are compared by using the same parameters. The results of optimal solutions and running times of three algorithms with different parameters are listed in Table VI.

From Table VI, the following conclusions can be obtained: First, the optimal solutions of the three algorithms are close, which shows that the hybrid intelligent algorithm is reasonable and feasible when it is used to solve an emergency scheduling problem for forest fires subject to priority disaster areas. Second, the results of each algorithm in the six experiments clearly show that the running time of PSO is shortest among them because PSO has good converge ability. Additionally, the optimal solutions of GA and the hybrid algorithm are basically consistent and able to obtain the optimal solutions effectively. However, after programs are run six times, the used average time of the hybrid algorithm is 704.617 ms, while the used average time of the GA algorithm is 3027.679 ms; especially, the operating efficiency was significantly reduced as the scale of $PopSize$ increased in GA. Hence, based on the results, the hybrid algorithm is feasible and effective to solve the proposed problem.

To further show the performance of the proposed approach, a comparison test is performed with different the number of fire points, i.e. 10, 15, and 20. Note that the distance between each point d_{ij} at least [50, 100], $PopSize = 50$, and $MaxIter = 100$. Other parameters are listed in Table VII, in which V_m is set to be 54 km/h and T'_m is adjusted with the change of the number of fire points, and the number of rescue teams M is changed when the number of fire points is the same. After the different algorithms were run, the results obtained are shown in Table VII.

It can be seen from Table VII that the quasi-optimal solution of the proposed algorithm is obviously superior to the other algorithms, and GA is inferior to PSO. PSO can search quasi-optimal solutions with competitive computational times, but its results are the worst among the three. Also, to demonstrate the importance of the limited rescue team resources in our model, we change the number of rescue teams M , i.e. $M = 3, 4$, and 5 , respectively. The objective function values become greater with increasing M , which shows that the amount of rescue resources has an impact on our problem. Thus, it is necessary to consider the limited resources of the rescue teams to emergency scheduling for forest fires. Thus, the experiments show the efficiency and

Table V. Comparison of solutions with different parameters

Times	PopSize	MaxIter	V_m	T'_m	Actual value	V^*	Relative error (%)	Running time (ms)
1	5	10	54	4	525	518	1.35%	40.943
2	10	20	54	4	521	518	0.58%	127.152
3	10	50	54	4	519	518	0.19%	474.064
4	20	20	54	4	518	518	0.00%	230.774
5	20	100	54	4	518	518	0.00%	1269.682
6	50	100	54	4	518	518	0.00%	2622.539

Table VI. Comparison of optimization results of hybrid intelligent algorithm with GA and PSO algorithms

PopSize	V_m	T'_m	MaxIter	Quasi-optimal solution			Running time (ms)		
				Proposed method	GA	PSO	The proposed method	GA	PSO
5	50	4	10	525	520	544	49.048	85.363	43.497
10			10	520	520	528	78.090	149.654	88.131
20			20	519	519	524	239.856	582.810	179.771
50			5	519	519	524	156.724	480.633	108.814
50			50	519	519	524	1254.220	4279.224	608.493
100			50	519	519	519	2449.762	12588.387	1079.208
Average value				520.1667	519.333	527.167	704.617	3027.679	351.319

Table VII. Comparison of optimization results of hybrid intelligent algorithm with GA and PSO algorithms with different fire points

Number of fire points	V_m	T'_m	M	Quasi-optimal solution			Running time (ms)		
				Proposed method	GA	PSO	The proposed method	GA	PSO
10	54	6	3	844.00	846.57	860.98	2608.09	7576.94	1800.71
			4	901.06	906.58	923.23	2699.49	7378.90	1825.97
			5	970.63	981.63	986.05	2856.33	7607.88	1899.61
15	54	9	3	1234.69	1253.24	1264.72	2334.46	6350.44	1779.14
			4	1332.17	1338.75	1345.24	2689.03	7008.38	1822.27
			5	1433.11	1445.48	1456.68	2858.79	7412.81	1932.94
20	54	12	3	1507.03	1516.70	1540.17	2231.39	6633.34	1802.01
			4	1602.01	1619.07	1625.10	2694.53	7037.78	1895.01
			5	1707.46	1728.40	1732.52	2872.48	7728.89	1961.46

effectiveness of the proposed algorithm in solving an emergency scheduling problem of forest fires subject to limited rescue team resources.

5. Conclusions

In this paper, we addressed, for the first time, an emergency scheduling problem for forest fire subject to limited rescue teams and priority disaster areas to minimize the travel time of the fire fighting teams. A hybrid intelligent algorithm integrating GA and PSO was adopted to solve the proposed problem. Compared with the GA and PSO, the hybrid algorithm could effectively solve an emergency scheduling problem of the forest fire. The results can be used to guide decision makers to take better decisions when a forest fire occurs.

In addition, for a more practically orientated application, a software package with a graphical user interface needs to be developed in future. Furthermore, a robust system capable of weighing as many factors as possible in the determination of feasible routing schedules needs to be studied.

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